

Dual groups for double Brylinski–Deligne covers

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Abstract

We introduce an explicit maximal-lattice deformation of the spherical dual root datum, depending on the first Brylinski–Deligne invariant modulo two. We prove that it is a root datum, give an exact Lie-generator criterion for a distinguished morphism to the covering dual.

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1 The problem and the meaning of a lift

Let k be a p -adic field, G a split connected reductive group, and $H \subset G$ a connected reductive spherical subgroup defined over k . Assume that the geometric spherical variety $X = H \backslash G$ has no type N spherical roots. Fix a BD extension of G and its Hilbert-symbol double cover

$$1 \longrightarrow \mu_2 \longrightarrow \tilde{G} \xrightarrow{p_G} G(k) \longrightarrow 1.$$

Let $\tilde{H} = p_G^{-1}(H(k))$. There is a canonical identification

$$\tilde{H} \backslash \tilde{G} \simeq H(k) \backslash G(k). \tag{1}$$

Indeed, send the class of g' to the class of $p_G(g')$. Surjectivity and injectivity follow directly from the definition of $\tilde{H}$. The right action of $\tilde{G}$ on this quotient factors through $G(k)$. In particular, for a genuine representation π ,

$$\mathrm{Hom}_{\tilde{H}}(\pi, \mathbf{1}) = 0 : \quad \ell(v) = \ell(\pi(-1)v) = -\ell(v).$$

Here $H(k) \backslash G(k)$ is the distinguished rational orbit; it should not automatically be identified with all of $X(k)$.

The ordinary theorem, Chen [1, Theorem 4.1.7] following Knop–Schalke [4], supplies a distinguished map

$$\iota_X : G_X^\vee \times \mathrm{SL}_2(\mathbb{C}) \longrightarrow G^\vee. \quad (2)$$

Restriction of the BD cover to H does not itself define a new spherical dual group or a comparison $\tilde{G}^\vee \rightarrow G^\vee$. For example, the basic double cover of Sp_6 has dual $\mathrm{Sp}_6(\mathbb{C})$, while the ordinary dual is $\mathrm{SO}_7(\mathbb{C})$. No nontrivial algebraic homomorphism $\mathrm{Sp}_6 \rightarrow \mathrm{SO}_7$ exists: a nonzero differential would be injective between simple Lie algebras of the same dimension 21, hence would incorrectly identify types C_3 and B_3 .

We therefore use *lift* in the precise deformation sense of Definition 3.1 below: deform the source root datum and the ambient root datum, retain the natural torus map and associated-root support, and use the principal SL_2 in the modified Levi. No nonexistent universal vertical map to $G^\vee$ is assumed. The resulting group $G_{X,Q}^{\vee, \max}$ is a proposed meaning of $G_X^\vee$ for this algebraic problem, not a claimed spectral dual of the scalar quotient (1).

2 A maximal-lattice root datum

Choose geometric tori and write

$$Y = X_*(T), \quad Y^* = X^*(T), \quad \Xi = X^*(T_X) \subset Y^*, \quad \Lambda = \Xi^*.$$

Restriction gives $p : Y \rightarrow \Lambda$. Its rational image is all of $\Lambda_{\mathbb{Q}}$, but its integral image need not be all of Λ . Write the ordinary spherical dual root datum as

$$(\Lambda, \Phi_X^\vee, \Xi, \Phi_X),$$

with simple spherical roots γ_i and spherical coroots $\beta_i = \gamma_i^\vee$. The normalization is the SV normalization used in the thesis. For type T, $\gamma_i = \alpha$ and $\beta_i = p(\alpha^\vee)$. For type G, fix its associated pair α, α' ; then

$$\gamma_i = \alpha + \alpha', \quad \beta_i = p(\alpha^\vee) = p(\alpha'^\vee).$$

The two associated ambient roots are superstrongly orthogonal. The associated-root data are those of the ordinary distinguished morphism, not an arbitrary decomposition of γ_i .

Let $Q : Y \rightarrow \mathbb{Z}$ be the Weyl-invariant first BD invariant and put

$$B_Q(y, z) = Q(y + z) - Q(y) - Q(z), \quad L = Y_{Q,2} = \{y \in Y : B_Q(y, Y) \subset 2\mathbb{Z}\},$$

$$n_\alpha = \frac{2}{\gcd(2, Q(\alpha^\vee))}.$$

The covering dual $K = \tilde{G}^\vee$ has root datum [3, §2.2]

$$(L, \{n_\alpha \alpha^\vee\}, L^*, \{\alpha/n_\alpha\}). \quad (3)$$

Define

$$d_i = \begin{cases} n_\alpha, & \gamma_i = \alpha \text{ of type T,} \\ \gcd(n_\alpha, n_{\alpha'}), & \gamma_i = \alpha + \alpha' \text{ of type G.} \end{cases} \quad b_i = d_i \beta_i, \quad c_i = \gamma_i / d_i. \quad (4)$$

Lemma 2.1. *Every b_i lies in $p(L)$, and every c_i is integral on $p(L)$. The lattice $p(L)$ is stable under the little Weyl group W_X .*

Proof. Weyl invariance gives

$$B_Q(\alpha^\vee, y) = Q(\alpha^\vee)\langle \alpha, y \rangle.$$

Consequently $n_\alpha \mid \langle \alpha, y \rangle$ for $y \in L$, and $n_\alpha \alpha^\vee \in L$. This proves the assertion for type T. For type G, choose integers u, v with $un_\alpha + vn_{\alpha'} = d_i$. Then

$$b_i = p(un_\alpha \alpha^\vee + vn_{\alpha'} \alpha'^\vee) \in p(L),$$

and $\langle \alpha + \alpha', y \rangle$ is divisible by d_i for $y \in L$. The spherical reflection is induced on $p(Y)$ by s_α in type T and by $s_\alpha s_{\alpha'}$ in type G. These preserve L , and generate W_X on $\Lambda_\mathbb{Q}$. $\square$

Theorem 2.2. *The multipliers d_i extend to a W_X -invariant multiplier d_γ on Φ_X . Put*

$$\begin{aligned} R_Q &= \{d_\gamma \gamma^\vee : \gamma \in \Phi_X\}, & C_Q &= \{\gamma/d_\gamma : \gamma \in \Phi_X\}, \\ M &= \{\lambda \in \Lambda : \langle c, \lambda \rangle \in \mathbb{Z} \text{ for every } c \in C_Q\}. \end{aligned} \tag{5}$$

Then (M, R_Q, M^, C_Q) is a reduced root datum. It has the same Weyl group as the ordinary spherical datum. Moreover,*

$$p(L) \subset M \subset \Lambda,$$

with finite indices. If $Q \equiv 0 \pmod{2}$ on Y , the datum is the ordinary spherical dual datum, including its full augmentation.

Proof. Lemma 2.1 implies integrality of

$$\langle c_i, b_j \rangle = \frac{d_j}{d_i} \langle \gamma_i, \beta_j \rangle.$$

Across a single Dynkin edge, both ordinary off-diagonal entries are -1 , so $d_i = d_j$. Simple roots in the same Weyl orbit are connected by single edges; hence the scaling extends invariantly. The resulting sets of scaled roots and coroots are stable under W_X . The reflection is unchanged:

$$\lambda \mapsto \lambda - \langle \gamma/d_\gamma, \lambda \rangle d_\gamma \gamma^\vee = \lambda - \langle \gamma, \lambda \rangle \gamma^\vee.$$

All roots lie in $p(L)$ by the lemma and Weyl stability, while all coroots are integral on M by definition. Their corresponding pairings are 2; all cross-pairings are integral. Reducedness follows from reducedness of the ordinary root system. These are the root-datum axioms. The finite-index assertions follow since p is rationally surjective and L has finite index in Y . For even Q , every multiplier is 1 and $M = \Lambda$. $\square$

Definition 2.3. The *maximal-lattice BD deformation* of the spherical dual is the pinned complex reductive group $G_{X,Q}^{\vee, \max}$ with the root datum in Theorem 2.2. The inclusion $p(L) \subset M$ defines its natural torus map

$$t_Q : T(G_{X,Q}^{\vee, \max}) \longrightarrow T(K), \quad t_Q^* = p|_L. \tag{6}$$

This map has finite kernel; it need not be an inclusion.

The maximal choice in (5) is a convention with a useful normalization. Taking only $p(L)$ can lose the ordinary augmentation even when $Q = 0$. More generally every W_X -stable lattice between $p(L)$ and M gives a root datum. The maximal choice recovers the original group in the even case. Its root datum records the relevant coroot parities; the torus map additionally records L . No claim is made that these data determine a genuine coefficient system, an L -group extension, or a Plancherel measure.

3 An exact distinguished-morphism criterion

Let S_X^p be the simple-root set of the ordinary spherical Levi. In K , use the Levi with modified simple roots $n_\alpha \alpha^\vee$ for $\alpha \in S_X^p$. Let

$$j_Q : \mathrm{SL}_2 \longrightarrow K$$

be its pinned principal homomorphism. Its semisimple differential element is the sum of the positive *modified* Levi coroots. One must not reuse the old principal cocharacter when multipliers change.

Definition 3.1. A *BD-distinguished deformation* is an algebraic homomorphism

$$F_Q : G_{X,Q}^{\vee, \max} \times \mathrm{SL}_2 \longrightarrow K$$

restricting to t_Q and j_Q , and satisfying the following support prescription. A simple source root space of type T maps nontrivially into the root space for $n_\alpha \alpha^\vee$. In type G it maps into the sum of the two associated modified root spaces, with a nonzero projection to each. The analogous prescription is imposed on negative simple root spaces.

Theorem 3.2. Let $h_i = dt_Q(c_i)$ and $a_{ij} = \langle c_i, b_j \rangle$. Write (e_0, f_0, h_0) for the differential triple of j_Q . A *BD-distinguished deformation* exists if and only if one can choose elements $E_i, F_i \in \mathrm{Lie}(K)$, in the prescribed positive and negative associated-root supports, satisfying:

$$\begin{aligned} \mathrm{Ad}(t_Q(s))E_i &= b_i(s)E_i, & \mathrm{Ad}(t_Q(s))F_i &= b_i(s)^{-1}F_i, \\ [E_i, F_j] &= \delta_{ij}h_i, \\ (\mathrm{ad} E_i)^{1-a_{ij}}E_j &= 0, & (\mathrm{ad} F_i)^{1-a_{ij}}F_j &= 0 \quad (i \neq j), \\ [E_i, e_0] &= [E_i, f_0] = [E_i, h_0] = 0, & [F_i, e_0] &= [F_i, f_0] = [F_i, h_0] = 0. \end{aligned}$$

Required associated-root coefficients must be nonzero. This is a finite algebraic criterion, including global central-isogeny descent.

Proof. Necessity follows from the Chevalley–Serre relations and commutation of the two source factors. Conversely the displayed equations, together with torus equivariance, give all Cartan and Chevalley–Serre relations and hence a homomorphism of derived Lie algebras. It integrates from the simply connected derived cover over $\mathbb{C}$. On its maximal torus this map agrees with t_Q , because the differentials agree and torus homomorphisms in characteristic zero are determined by their differentials. It therefore kills the kernel of the simply connected cover of the derived group. Extend by t_Q on the connected center, and descend the finite central kernel of the product of that center with the simply connected derived group; again the required agreement holds on the product maximal torus. This gives a map from the exact group $G_{X,Q}^{\vee, \max}$.

For $\alpha \in S_X^p$, $p(\alpha^\vee) = 0$, so the modified Levi roots restrict trivially under t_Q . Thus the source torus centralizes the derived modified Levi. The final equations imply that the two connected group images commute, giving F_Q on the product. Choosing target Chevalley bases turns the equations into finitely many polynomial equations; nonzero coefficients can be imposed with auxiliary inverses. $\square$

Proposition 3.3. For a type G associated pair α, α' , existence requires $n_\alpha = n_{\alpha'}$.

Proof. The two target root spaces have source torus weights $n_\alpha \beta$ and $n_{\alpha'} \beta$, with $\beta \neq 0$. An eigenvector cannot have nonzero components of two distinct weights. $\square$

Proposition 3.4. *If n_α is one common value d on all ambient roots, a BD-distinguished deformation exists. In particular this holds for simple ambient types A, D, E, G_2 for either BD-level parity, and for every ambient group when Q is even.*

Proof. The source and target root systems are uniform rescalings of the ordinary systems. Identify root vectors and divide toral coroots by d . Transport the ordinary distinguished Lie map through these identifications. Its torus differential remains dt_Q because source and target have the same scaling. It preserves the required support and carries the ordinary principal triple to the modified principal triple. Theorem 3.2 integrates it on the full lattice M . In type G_2 the normalized coroot values are $a, 3a$, so their parities agree. $\square$

4 Simple ambient groups: every Krämer entry

This section applies to the *actual connected subgroup* in the simply connected ambient group. Orthogonal matrix notation in classification tables is interpreted through the appropriate subgroup or connected inverse image in Spin. Write $Q = aq_G$, where q_G is 1 on short coroots.

Theorem 4.1. *For every proper connected reductive spherical subgroup in Krämer’s list satisfying the no-type-N hypothesis, the proposed BD-distinguished deformation exists as follows:*

- If a is even, it exists in every case.
- If a is odd, it exists in types A, D, E, G_2 and fails in types B_r, C_r ($r \geq 2$) and F_4 .

Rank one $B_1 = C_1 = A_1$ is included in type A . If the trivial pair $H = G$ is added, it always works.

The following exhaustive tables distinguish an obstruction from exclusion by the premise. “N” means that the pair has type N and is outside the definition used here. “Yes” and “No” refer to odd a ; every eligible row works for even a . Usual valid rank bounds and the familiar low-rank identifications are understood.

G	H or its indicated image/preimage	Odd-level outcome
SL_d	SO_d	Yes for $d = 2$; N for $d \geq 3$
SL_{p+q}	$S(GL_p \times GL_q)$	Yes
SL_{p+q}	$SL_p \times SL_q$, $p > q \geq 1$	Yes
SL_{2r}	Sp_{2r}	Yes
SL_{2r+1}	$G_m \cdot Sp_{2r}$	Yes
SL_{2r+1}	Sp_{2r}	Yes
Sp_{2r}	GL_r	Yes for $r = 1$; N for $r \geq 2$
Sp_{2r}	$G_m \times Sp_{2r-2}$, $r \geq 2$	No: bracket obstruction
Sp_{2p+2q}	$Sp_{2p} \times Sp_{2q}$, $p, q \geq 1$	No: Arthur obstruction
$Spin_{2r}$	Preimage of GL_r	Yes
$Spin_{2r}$	SL_r , r odd	Yes
$Spin_8$	Image/preimage of $Sp_4 \otimes Sp_2$	N
$Spin_8$	$Spin_7$	Yes
$Spin_8$	G_2	Yes
$Spin_{10}$	Preimage of $SO_2 \times Spin_7$	Yes
$Spin_{2r+1}$	Preimage of GL_r , $r \geq 2$	No: bracket obstruction
$Spin_N$	Preimage of $SO_a \times SO_b$, $a \geq b$, $a + b = N$	Even N : Yes if $b \leq 2$, otherwise N. Odd N : No if $b = 1$, otherwise N.

G	H or its indicated image/preimage	Odd-level outcome
Spin_9	Spin_7	No: associated-weight obstruction
Spin_7	G_2	No: associated-weight obstruction

G	Type of H	Odd-level outcome
G_2	A_2	Yes
G_2	$A_1 \times A_1$	N
F_4	B_4	No: Arthur obstruction
F_4	$C_3 \times A_1$	N
E_6	C_4	N
E_6	F_4	Yes
E_6	D_5	Yes
E_6	$\mathbf{G}_m \cdot D_5$	Yes
E_6	$A_5 \times A_1$	N
E_7	$\mathbf{G}_m \cdot E_6$	Yes
E_7	A_7	N
E_7	$D_6 \times A_1$	N
E_8	D_8	N
E_8	$E_7 \times A_1$	N

4.1 Proofs of the negative rows

For $G = \text{Sp}_{2r}$ and odd a , take standard coordinates. The covering dual has roots

$$\{\pm e_i \pm e_j\} \cup \{\pm 2e_i\},$$

whereas the ordinary dual has roots $\pm e_i \pm e_j, \pm e_i$. For $\mathbf{G}_m \times \text{Sp}_{2r-2}$ the normalized spherical roots $e_1 - e_2$ and $e_1 + e_2$ are orthogonal. Their forced target root vectors do not commute: their sum $2e_1$ is a root. A Chevalley bracket in that root space is nonzero, contradicting the source relation.

For $\text{Sp}_{2p} \times \text{Sp}_{2q}$, $p \leq q$, the last normalized spherical root is

$$\sigma = e_{2p-1} + e_{2p}.$$

The spherical Levi includes $\alpha_{2p-1} = e_{2p-1} - e_{2p}$. The corresponding forced root vector and the relevant summand of the principal Levi nilpotent have a nonzero bracket of weight $2e_{2p-1}$. Other Levi summands have distinct resulting weights, so cancellation is impossible. This proves the Arthur obstruction, including $p = q = 1$.

For ambient B_r at odd level the modified dual roots are $2B_r$ in the usual coordinates. In the sphere case $\text{SO}_{2r} \subset \text{SO}_{2r+1}$, the spherical root is e_1 , and the Levi contains its terminal simple root e_r . The forced target root vectors of weights $2e_1$ and $2e_r$ have nonzero bracket of weight $2(e_1 + e_r)$. For $\text{GL}_r \subset \text{SO}_{2r+1}$, the last two normalized spherical roots are e_{r-1} and e_r ; their orthogonal source root spaces must commute, whereas their target vectors do not.

For $G_2 \subset \text{SO}_7$, the type G spherical root $e_1 + e_2 + e_3$ has associated roots $e_1 + e_3$ and e_2 , of different lengths. Their BD multipliers are 2 and 1. Proposition 3.3 applies. The $\text{Spin}_7 \subset \text{SO}_9$ row contains the same type B_3 configuration on coordinates 2, 3, 4.

For $B_4 \subset F_4$, use Bourbaki coordinates

$$\alpha_1 = e_2 - e_3, \quad \alpha_2 = e_3 - e_4, \quad \alpha_3 = e_4, \quad \alpha_4 = (e_1 - e_2 - e_3 - e_4)/2.$$

The spherical root is $\sigma = e_1$ and $S_X^p = \{\alpha_1, \alpha_2, \alpha_3\}$. The odd-level modified dual has root set $2F_4$. The forced spherical root vector of weight $2e_1$ fails to commute with the Levi simple-root vector of weight $2e_4$: their sum $2(e_1 + e_4)$ is a root. Again the remaining principal-nilpotent summands cannot cancel this weight component.

These proofs exhaust the eligible nonuniform cases. The positive cases follow from Proposition 3.4, proving Theorem 4.1. The type-N filter comes from the connected-subgroup augmentations of the systems in Bravi–Pezzini, not from blindly using the normalizer lattice. For example, $\mathrm{SO}_d \subset \mathrm{SL}_d$ has doubled simple roots for $d \geq 3$, while the connected torus in SL_2 has the undoubled root. In even orthogonal block cases, the terminal $2e_b$ is not twice a root of D_r ; in odd orthogonal cases it is twice a root of B_r . This explains the different cutoffs in the table.

Here are explicit checks of the potentially delicate augmentations. For $\mathrm{SL}_d/\mathrm{SO}_d$ and $\mathrm{Sp}_{2r}/\mathrm{GL}_r$, the connected-subgroup weight lattice is $2P$, with P the ambient weight lattice. For $d \geq 3$, every simple root of A_{d-1} has an odd fundamental-weight coefficient; in C_r , this holds for the first $r - 1$ simple roots. They therefore remain type N. For E_6/C_4 , the connected augmentation of the displayed Bravi–Pezzini subgroup has index 3; an odd-index enlargement of its lattice $2Q(E_6)$ cannot contain a simple root. For E_7/A_7 , the enlargement has index 2, whereas the seven simple-root classes modulo $2Q(E_7)$ are independent over $\mathbf{F}_2$; at least six doubled directions remain type N. The disconnected orthogonal block normalizers also have component group of order 2. In the excluded ranges their lattices have at least two independent type-N classes, so passing to the connected subgroup cannot remove all of them. The connected odd-orthogonal block with $b = 2$ is itself listed with a doubled terminal root. These checks also distinguish the connected subgroups from their normalizers.

5 Product groups: cases (2)–(9) of Theorem 1.1.28

Use the exact group embeddings, with the qualifications in Section 6. Normalize each simple ambient factor by $q(\text{short coroot}) = 1$ and write its integral level as a, b, c in the order of the thesis. The conditions below depend only on these integers modulo two. An absent rank-zero factor has no level. For a semisimple factor such as $\mathrm{Spin}_4 = \mathrm{SL}_2^2$, its two simple factors may have independent levels.

Theorem 5.1. *For the spherical pairs in cases (2)–(9), the BD-distinguished deformation of Definition 3.1 exists exactly under the conditions in the following table, on the valid spherical parameter ranges.*

Case	Ambient group / range	Necessary and sufficient condition
2	$J \times J$, diagonal simple J	$a \equiv b \pmod{2}$
3	SL_2^3 , diagonal SL_2	All levels allowed
4a	$\mathrm{SL}_2 \times \mathrm{Sp}_{2n+2}$, $n \geq 1$	a, b even
4b	$\mathrm{Sp}_4 \times \mathrm{Sp}_{2n+4}$, $n = 1$	All levels allowed
4b	Same, $n \geq 2$	b even; a arbitrary
4c	$\mathrm{Spin}_n \times \mathrm{Spin}_{n+1}$, orthogonal graph, $n \geq 3$	All simple-factor levels allowed
4d	$\mathrm{Spin}_7 \times \mathrm{Spin}_8$, spin graph	All levels allowed
5	$\mathrm{Sp}_{2m+2} \times \mathrm{Sp}_{2n+2}$, at least one positive tail	a, b even
6	$\mathrm{Sp}_4 \times \mathrm{Sp}_{2m+2} \times \mathrm{Sp}_{2n+2}$	b even if $m \geq 1$; c even if $n \geq 1$; a arbitrary

Case	Ambient group / range	Necessary and sufficient condition
7	$\mathrm{Sp}_{2\ell+2} \times \mathrm{Sp}_{2m+2} \times \mathrm{Sp}_{2n+2}$	Each factor of rank ≥ 2 has even level; rank-one factors unrestricted
8	$\mathrm{SL}_n \times \mathrm{SL}_{n+1}$	All levels allowed
9	$\mathrm{SL}_{m+2} \times \mathrm{Sp}_{2n+2}$, $n \geq 1$	b even; a arbitrary
9	Same, $n = 0$	All levels allowed

In case (9), the variant with the torus has $m \geq 1$; the variant without the torus is spherical only for $m \geq 3$. The table applies to both on these ranges. In (4a) and (4b), the formal boundary $n = 0$ is a diagonal group case and is governed by (2). In (5), $m = n = 0$ is also case (2). In (6), $m = n = 0$ is allowed with no parity restriction. In (7), three zero tails give case (3). Rank-one and other duplicate presentations are interpreted by their actual root data.

5.1 Root configurations and proof

The configurations used here are read from Bravi–Pezzini [5, §3, entries 42–50], with the actual connected-subgroup augmentation retained. Diagram lines identifying colors must not be mistaken for sums of spherical roots.

Case (2). Each spherical root has an associated pair consisting of corresponding roots in the two ambient factors. If the levels have different parities, a short coroot gives multipliers 1 and 2; Proposition 3.3 forbids a distinguished map. If the parities agree, the two covering dual data coincide. The candidate is exactly that covering dual, and the diagonal map works (up to the harmless root-vector signs in a chosen distinguished normalization). To check the full lattice, here $\Lambda = Y_J$ and

$$M = \{y \in Y_J : n_\alpha \mid \langle \alpha, y \rangle \ \forall \alpha\} = Y_{J,Q,2},$$

because the simple coroots form a basis of Y_J and the identity in Lemma 2.1 detects all pairings with that basis.

Cases (3), (4c), (4d), and (8). Every ambient simple root is a normalized spherical root, and $S_X^p = \emptyset$. Thus the modified spherical root system is the full modified ambient root system. The root-vector assignment is the identity on every simple component; the maximal lattice supplies the appropriate finite central cover, and Theorem 3.2 integrates it. There is no Arthur commutation condition beyond the trivial Levi. Triality gives (4d) from the corresponding orthogonal graph case. This proves existence for independent parities, rather than requiring the ambient levels to agree.

For illustration, in (3)

$$\Xi = \{u\omega_1 + v\omega_2 + w\omega_3 : u + v + w \equiv 0 \pmod{2}\}, \quad \Lambda = \mathbb{Z}^3 + \mathbb{Z}(1/2, 1/2, 1/2).$$

The maximal lattice equals Λ if all levels are even, and equals $\mathbb{Z}^3$ otherwise. This explicitly retains the finite augmentation which the image lattice $p(L)$ would lose in the even case.

Case (4a). For $r = n + 1 \geq 2$, the spherical roots are

$$\gamma_0 = \alpha_0 + \alpha_1, \quad \gamma_1 = \alpha_1 + 2 \sum_{i=2}^{r-1} \alpha_i + \alpha_r = e_1 + e_2,$$

where α_0 belongs to the external SL_2 and $\alpha_1 = e_1 - e_2$ to Sp_{2r} . They form an A_1^2 system. The symplectic short-root multiplier for α_1 is always 1; hence a must be even by the associated-pair weight test. If b is odd, the symplectic component of the γ_0 root vector brackets nontrivially with the γ_1 vector, in weight $2e_1$. No component in the external factor can cancel it. Thus b must also be even. When both are even the ordinary map supplies the required deformation.

Case (5). With both tails positive the spheres are

$$\alpha_1 + \alpha'_1, \quad e_1 + e_2, \quad f_1 + f_2.$$

They form A_1^3 . Either odd ambient level gives the same nonzero forbidden bracket in that factor. Both even levels suffice. With exactly one zero tail this is (4a), and with both zero tails this is (2).

Cases (6) and (7). Each positive symplectic tail contributes the two type T roots $e_1 - e_2, e_1 + e_2$ within its own factor; a rank-one factor contributes its unique simple root. The central Sp_4 of (6) contributes its *full* simple root system. Thus odd level is forbidden precisely on the partial symplectic factors with a positive tail. If their levels are even, the roots $e_1 \pm e_2$ are a closed D_2 subsystem in the ambient B_r dual. The remaining Levi has coordinates $3, \dots, r$, so its principal SL_2 centralizes this subsystem. Full ambient factors and rank-one factors work at either parity. Products of these assignments prove the asserted sufficiency.

Case (4b). For $n = 1$, all simple roots in $C_2 \times C_3$ are spherical, so the full-system argument applies. For $n \geq 2$, put $r = n + 2$. The first Sp_4 contributes its full system. The second factor contributes

$$\alpha_1, \quad \alpha_2, \quad \alpha_3, \quad \alpha_3 + 2 \sum_{i=4}^{r-1} \alpha_i + \alpha_r = e_3 + e_4.$$

Their associated roots form D_4 on the first four coordinates. If b is odd, the root vectors for $e_3 - e_4$ and $e_3 + e_4$ have a nonzero bracket of weight $2e_3$, contrary to their source relation. If b is even, this is a closed D_4 subgroup in B_r , centralized by the complementary Levi on coordinates $5, \dots, r$. The first-factor full system puts no condition on a .

Case (9). The type A factor contributes, for $m \geq 2$,

$$\alpha_1, \quad \alpha_{m+1}, \quad \delta = \alpha_2 + \dots + \alpha_m = e_2 - e_{m+1}.$$

These form a closed A_3 subsystem on the four coordinates $1, 2, m+1, m+2$. For $m = 1$ it is simply the full A_2 system. Uniform modification in type A preserves its Lie embedding; the complementary Levi acts on coordinates $3, \dots, m$. The symplectic factor contributes $e_1 \pm e_2$ when $n \geq 1$, forcing and, by the closed- D_2 construction, allowing exactly even b . When $n = 0$ it contributes a full A_1 factor. Removing the torus from H in its valid spherical range changes the augmentation and adds a central torus on the spherical-dual side, but does not change these root or Levi obstructions. The torus map and the integration argument handle this change.

These arguments prove both directions in every row of Theorem 5.1. A negative result concerns Definition 3.1; it is not a claim that every unrelated algebraic homomorphism to K is impossible.

5.2 The basic double cover in each factor

For quick reference, put every nontrivial simple-factor level equal to 1.

- Case (1) is governed by Theorem 4.1, with type-N rows excluded.
- Cases (2), (3), (4c), (4d), and (8) work.
- Case (4a) fails for $n \geq 1$. Case (4b) works for $n = 1$ and fails for $n \geq 2$.
- Case (5) fails if there is any positive tail. Case (6) works only when both external tails vanish. Case (7) works only when all three tails vanish.
- Case (9) fails for $n \geq 1$ and works for $n = 0$, on its valid spherical parameter ranges.

These statements concern the BD double cover even when the restriction to H is split. For example in (2) equal odd ambient levels restrict to an even level on H , but the proposed spherical dual is the odd-level covering dual of J , not the dual of the split restriction to H .

6 Group forms and parameter corrections

Brion explicitly identifies isogenous groups in the classification [6, p. 191]. This convention must be resolved for BD root data.

Orthogonal entries. The literal SO factors in case (4) conflict with the simply connected hypothesis in the theorem. Use the corresponding Spin graph embeddings and actual connected subgroups. The tables above use that convention. They do not silently assign a simply connected cocharacter lattice to a literal matrix SO group. For any other chosen isogeny form, the construction and exact criterion still apply on its actual lattices, but the simply connected level tables are not a substitute for that calculation.

The torus parameter in case (9). The displayed homomorphism

$$(u, v, w, \lambda) \mapsto ((\lambda^m u) \oplus (\lambda^{-2} v), u \oplus w)$$

has kernel

$$\{(I_2, \lambda^2 I_m, I_{2n}, \lambda) : \lambda^m = 1\} \simeq \mu_m.$$

Thus the actual subgroup is the quotient of the displayed product by this kernel. Its full augmentation must be used in Ξ ; the maximal-lattice construction and torus descent keep track of it. This finite kernel does not change the Lie-theoretic parity obstructions above.

Removing the torus in case (9). The semisimple variant is spherical only for $m \geq 3$ (with $n \geq 0$). The missing small-rank restriction cannot be ignored. For $m = 2$, the irreducible representation $\bigwedge^2 k^4 \boxtimes \mathbf{1}$ has two H -fixed lines, namely $\bigwedge^2 U$ and $\bigwedge^2 V$ for the two two-dimensional special-linear blocks. For $m = 1$, use the irreducible representation $\mathfrak{sl}_3 \boxtimes k^{2n+2}$. On the common SL_2 ,

$$\mathfrak{sl}_3 \simeq \mathrm{Sym}^2 U \oplus U \oplus U \oplus \mathbf{1}, \quad k^{2n+2} \simeq U \oplus W,$$

so there are again two independent H -fixed vectors. Multiplicity-freeness therefore fails for every n , not just the values detected by a dimension count. For $m \geq 3$, the usual parabolic reduction

gives the open-orbit test for $T_{\mathrm{SL}_2} \times B_{\mathrm{SL}_m}$ acting on $2 \times m$ matrices. The remaining $m - 2$ diagonal coordinates absorb the determinant condition, so the projection to the first two columns is the B_{GL_2} action; together with the one-dimensional row torus this has an open orbit on the invertible 2×2 block. The remaining columns are cleared by the upper triangular unipotent action. This proves the stated spherical range. With the torus included, the small values $m = 1, 2$ are spherical as well.

7 What is proved, and what remains a spectral question

Theorems 2.2 and 3.2 give a precise proposed group and an exact existence test. Theorems 4.1 and 5.1 apply it to all pairs in the referenced classification that satisfy the geometric no-type-N hypothesis, distinguishing excluded parameters and group-form issues from failed distinguished deformations. The results are about connected complex dual groups. They do not claim an extension to full L -groups or independence from extra data in a genuine relative theory.

The full-preimage quotient in (1) does not distinguish genuine representations. A genuine model would require additional data, for example a splitting subgroup or genuine coefficients under $\tilde{H}$. In the group case with identical double-cover data on two factors, a splitting can be written explicitly:

$$\tilde{G} = (\tilde{J} \times \tilde{J}) / \{(z, z^{-1}) : z \in \mu_2\}, \quad s(h) = [\tilde{h}, \tilde{h}].$$

It is independent of the lift because $z^2 = 1$, and

$$s(J(k)) \backslash \tilde{G} \simeq \tilde{J}, \quad [g_1, g_2] \mapsto g_1^{-1} g_2.$$

The diagonal map of covering dual groups then has the expected group-case interpretation. This is a different quotient from the full-preimage quotient. Establishing comparable spectral interpretations for the other positive rows requires such additional relative data and further analysis; it is not a consequence of the root-datum construction.

References

- [1] R. Chen, *Spherical functions on spherical varieties for unramified groups*, supplied thesis dated 8 October 2026. Theorem 1.1.28, pp. 20–21; Proposition 4.1.5, Definition 4.1.6, and Theorem 4.1.7, pp. 142–143.
- [2] J.-L. Brylinski and P. Deligne, *Central extensions of reductive groups by K_2* , Publ. Math. IHES 94 (2001), 5–85. <https://www.numdam.org/item/10.1007/s10240-001-8192-2.pdf>.
- [3] M. H. Weissman, *L-groups and parameters for covering groups*, in *L-groups and the Langlands program for covering groups*, Asterisque 398 (2018), 33–186. Especially §1.2, §2.2, and §2.7. Included in the supplied covering-groups volume. <https://arxiv.org/abs/1507.01042>.
- [4] F. Knop and B. Schalke, *The dual group of a spherical variety*, Trans. Moscow Math. Soc. 78 (2017), 187–216. Associated roots and folding: §§6–7; centralizers and principal SL_2 : §9. <https://arxiv.org/abs/1702.08264>.
- [5] P. Bravi and G. Pezzini, *The spherical systems of the wonderful reductive subgroups*, J. Lie Theory 25 (2015), 105–123; arXiv revision v2 (2017), §3, entries 1–50. Use connected-subgroup augmentations as in §2. <https://arxiv.org/abs/1109.6777>.
- [6] M. Brion, *Classification des espaces homogenes spheriques*, Compositio Math. 63 (1987), 189–208. https://www.numdam.org/item/CM_1987__63_2_189_0.pdf.